Fermi arc model

standard BCS wave function

$$|\psi\rangle = \prod_k \left(u_k + v_k b_k^\dagger\right)|0\rangle$$

$b_k^\dagger = \hat{c}_{k\uparrow}^\dagger \hat{c}_{-k\downarrow}^\dagger$

appendix:

$$\langle\psi|n_{k\sigma}|\psi\rangle = |v_k|^2$$

$$H = \sum_{k\sigma}\left(\xi_k n_{k\sigma} + \frac{U}{2} n_{k\sigma} n_{k+Q-\sigma}\right) - \frac{J}{N}\sum_{k,k'}\gamma_{k,k'} c_{k\uparrow}^\dagger c_{-k,\downarrow}^\dagger c_{-k',\downarrow} c_{k',\uparrow}$$

Sum of spin

$\sum_\sigma \frac{U}{2} n_{k\sigma} n_{k+Q-\sigma} = U n_k n_{k+Q}$

$$\begin{aligned}\langle\psi|H|\psi\rangle = &\sum_k 2\xi_k|v_k|^2 + U|v_k|^2|v_{k+Q}|^2 \\ &- \frac{J}{N}\sum_{k,k'}\gamma_{k,k'} u_k v_k^* u_{k'}^* v_{k'}\end{aligned}$$

change variable

$$u_k = \cos\theta_k \quad v_k = \sin\theta_k$$

$$\begin{aligned}\langle\psi|H|\psi\rangle &= \sum_k\left[2\xi_k\sin^2\theta_k + U\sin^2\theta_k\sin^2\theta_{k+Q}\right] - \frac{J}{4N}\sum_{k,k'}\gamma_{k,k'}\sin 2\theta_k \sin 2\theta_{k'} \\ &= \sum_k\left[\xi_k(1-\cos 2\theta_k) + U\sin^2\theta_k\sin^2\theta_{k+Q}\right] - \frac{J}{4N}\sum_{k,k'}\gamma_{k,k'}\sin 2\theta_k\sin 2\theta_{k'}\end{aligned}$$

$\gamma_{k,k'} = \gamma_k \times \gamma_{k'}$

Second term:

$$\begin{aligned}&\frac{\partial}{\partial\theta_k}\sum_p U\sin^2\theta_p\sin^2\theta_{p+Q} \\ =&\frac{\partial}{\partial\theta_k}U\left(\sin^2\theta_k\sin^2\theta_{k+Q} + \sin^2\theta_{k-Q}\sin^2\theta_k\right) \\ =&U\sin 2\theta_k\sin^2\theta_{k+Q} + U\sin 2\theta_k\sin^2\theta_{k-Q}\end{aligned}$$

$\sin^2\theta_{k-Q} = \sin^2\theta_{k-Q+G} = \sin^2\theta_{k+Q}$

Variational principle:

$$\frac{\partial}{\partial\theta_k}E = 2\xi_k\sin 2\theta_k + 2U\sin 2\theta_k\sin^2\theta_{k+Q} - \frac{J}{N}\gamma_k\cos 2\theta_k\sum_{k'}\gamma_{k'}\sin 2\theta_{k'} = 0$$

Gap:

$$\Delta = \frac{J}{2N}\langle\hat{\Delta}^\dagger\rangle = \frac{J}{2N}\sum_{\mathbf{k}}\gamma_{\mathbf{k}}\sin 2\theta_{\mathbf{k}}.$$

Appendix

1. $\langle\psi|n_{k\sigma}|\psi\rangle = |v_k|^2$

$$\begin{aligned}
\langle T_k\rangle =&\langle\Psi_\mathrm{G}\mid \sum_\sigma \xi_k c_{k\sigma}^\dagger c_{k\sigma}\mid\Psi_\mathrm{G}\rangle\\
=&\sum_\sigma \xi_k\langle 0|\left(u_{-\mathbf{k}}^*+v_{-\mathbf{k}}^*\hat c_{\mathbf{k}\downarrow}\hat c_{-\mathbf{k}\uparrow}\right)\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat c_{-\mathbf{k}\downarrow}\hat c_{\mathbf{k}\uparrow}\right)|c_{k\sigma}^\dagger c_{k\sigma}|\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)\left(u_{-\mathbf{k}}+v_{-\mathbf{k}}\hat c_{-\mathbf{k}\uparrow}^\dagger\hat c_{\mathbf{k}\downarrow}^\dagger\right)|0\rangle\times\\
&\prod_{\mathbf{k}'\neq\mathbf{k},\mathbf{k}'\neq-\mathbf{k}}\langle 0|\left(u_{\mathbf{k}'}^*+v_{\mathbf{k}'}^*\hat c_{-\mathbf{k}'\downarrow}\hat c_{\mathbf{k}'\uparrow}\right)|\left(u_{\mathbf{k}'}+v_{\mathbf{k}'}\hat c_{\mathbf{k}'\uparrow}^\dagger\hat c_{-\mathbf{k}'\downarrow}^\dagger\right)|0\rangle\\
=&\xi_k\langle 0|\left(u_{-\mathbf{k}}^*+v_{-\mathbf{k}}^*\hat c_{\mathbf{k}\downarrow}\hat c_{-\mathbf{k}\uparrow}\right)|c_{k\downarrow}^\dagger c_{k\downarrow}|\left(u_{-\mathbf{k}}+v_{-\mathbf{k}}\hat c_{-\mathbf{k}\uparrow}^\dagger\hat c_{\mathbf{k}\downarrow}^\dagger\right)|0\rangle+\xi_k\langle 0|\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat c_{-\mathbf{k}\downarrow}\hat c_{\mathbf{k}\uparrow}\right)|c_{k\uparrow}^\dagger c_{k\uparrow}|\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)|0\rangle\\
=&2\xi_k|v_\mathbf{k}|^2
\end{aligned}$$

2. BCS theory

$$|\Psi_\mathrm{G}\rangle=\prod_\mathbf{k}\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)|0\rangle$$

$$H=\sum_{k\sigma}(\epsilon_k-\mu)c_{k\sigma}^\dagger c_{k\sigma}-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}c_{k\uparrow}^\dagger c_{-k,\downarrow}^\dagger c_{-k',\downarrow}c_{k',\uparrow}$$

Normalization condition

$$\langle 0|\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat c_{-\mathbf{k}\downarrow}\hat c_{\mathbf{k}\uparrow}\right)|\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)|0\rangle=|u_\mathbf{k}|^2+|v_\mathbf{k}|^2=1$$

$$\langle 0|\left(u_{\mathbf{k}'}^*+v_{\mathbf{k}'}^*\hat c_{-\mathbf{k}'\downarrow}\hat c_{\mathbf{k}'\uparrow}\right)|\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)|0\rangle=0$$

kinetic term

$$\begin{aligned}
\langle T_k\rangle =&\langle\Psi_\mathrm{G}\mid \sum_\sigma \xi_k c_{k\sigma}^\dagger c_{k\sigma}\mid\Psi_\mathrm{G}\rangle\\
=&\sum_\sigma \xi_k\langle 0|\left(u_{-\mathbf{k}}^*+v_{-\mathbf{k}}^*\hat c_{\mathbf{k}\downarrow}\hat c_{-\mathbf{k}\uparrow}\right)\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat c_{-\mathbf{k}\downarrow}\hat c_{\mathbf{k}\uparrow}\right)|c_{k\sigma}^\dagger c_{k\sigma}|\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)\left(u_{-\mathbf{k}}+v_{-\mathbf{k}}\hat c_{-\mathbf{k}\uparrow}^\dagger\hat c_{\mathbf{k}\downarrow}^\dagger\right)|0\rangle\times\\
&\prod_{\mathbf{k}'\neq\mathbf{k},\mathbf{k}'\neq-\mathbf{k}}\langle 0|\left(u_{\mathbf{k}'}^*+v_{\mathbf{k}'}^*\hat c_{-\mathbf{k}'\downarrow}\hat c_{\mathbf{k}'\uparrow}\right)|\left(u_{\mathbf{k}'}+v_{\mathbf{k}'}\hat c_{\mathbf{k}'\uparrow}^\dagger\hat c_{-\mathbf{k}'\downarrow}^\dagger\right)|0\rangle\\
=&\xi_k\langle 0|\left(u_{-\mathbf{k}}^*+v_{-\mathbf{k}}^*\hat c_{\mathbf{k}\downarrow}\hat c_{-\mathbf{k}\uparrow}\right)|c_{k\downarrow}^\dagger c_{k\downarrow}|\left(u_{-\mathbf{k}}+v_{-\mathbf{k}}\hat c_{-\mathbf{k}\uparrow}^\dagger\hat c_{\mathbf{k}\downarrow}^\dagger\right)|0\rangle+\xi_k\langle 0|\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat c_{-\mathbf{k}\downarrow}\hat c_{\mathbf{k}\uparrow}\right)|c_{k\uparrow}^\dagger c_{k\uparrow}|\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)|0\rangle\\
=&2\xi_k|v_\mathbf{k}|^2
\end{aligned}$$

Inversion symmetry $v_\mathbf{k}=v_{-\mathbf{k}}$

Total kinetic term

$$\langle T\rangle=\sum_k\langle T_k\rangle=2\sum_k\xi_k|v_\mathbf{k}|^2$$

Coulomb interaction term (omit $k=k'$ term)

$$\begin{aligned}
\langle V\rangle =&\langle\Psi_\mathrm{G}\mid -\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}c_{k,\uparrow}^\dagger c_{-k,\downarrow}^\dagger c_{-k',\downarrow}c_{k',\uparrow}\mid\Psi_\mathrm{G}\rangle\\
=&-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}\langle 0\mid\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat c_{-\mathbf{k}\downarrow}\hat c_{\mathbf{k}\uparrow}\right)\left(u_{\mathbf{k}'}^*+v_{\mathbf{k}'}^*\hat c_{-\mathbf{k}'\downarrow}\hat c_{\mathbf{k}'\uparrow}\right)\mid c_{k\uparrow}^\dagger c_{-k,\downarrow}^\dagger c_{-k',\downarrow}c_{k',\uparrow}\mid\left(u_{\mathbf{k}'}+v_{\mathbf{k}'}\hat c_{\mathbf{k}'\uparrow}^\dagger\hat c_{-\mathbf{k}'\downarrow}^\dagger\right)\left(u_\mathbf{k}+v_\mathbf{k}\hat c_{\mathbf{k}\uparrow}^\dagger\hat c_{-\mathbf{k}\downarrow}^\dagger\right)\mid 0\rangle\\
=&-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}\langle 0\mid\left(u_\mathbf{k}^*+v_\mathbf{k}^*\hat b_\mathbf{k}\right)\left(u_{\mathbf{k}'}^*+v_{\mathbf{k}'}^*\hat b_{\mathbf{k}'}\right)\mid\hat b_k^\dagger\hat b_{k'}\mid\left(u_{\mathbf{k}'}+v_{\mathbf{k}'}\hat b_{\mathbf{k}'}^\dagger\right)\left(u_\mathbf{k}+v_\mathbf{k}\hat b_\mathbf{k}^\dagger\right)\mid 0\rangle\\
=&-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}\langle 0\mid\left(u_\mathbf{k}^*u_{\mathbf{k}'}^*+u_\mathbf{k}^*v_{\mathbf{k}'}^*\hat b_{\mathbf{k}'}+v_\mathbf{k}^*u_{\mathbf{k}'}^*\hat b_\mathbf{k}+v_\mathbf{k}^*v_{\mathbf{k}'}^*\hat b_\mathbf{k}\hat b_{\mathbf{k}'}\right)\mid\hat b_k^\dagger\hat b_{k'}\mid\left(u_\mathbf{k}u_{\mathbf{k}'}+u_\mathbf{k}v_{\mathbf{k}'}\hat b_{\mathbf{k}'}^\dagger+v_\mathbf{k}u_{\mathbf{k}'}\hat b_\mathbf{k}^\dagger+v_\mathbf{k}v_{\mathbf{k}'}\hat b_{\mathbf{k}'}^\dagger\hat b_\mathbf{k}^\dagger\right)\mid 0\rangle\\
=&-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}v_\mathbf{k}^*u_{\mathbf{k}'}^*u_\mathbf{k}v_{\mathbf{k}'}\\
=&-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}u_\mathbf{k}v_\mathbf{k}^*u_{\mathbf{k}'}^*v_{\mathbf{k}'}
\end{aligned}$$

$$E=2\sum_k\xi_k|v_\mathbf{k}|^2-\frac{g}{V}\sum_{k,k'}\gamma_{k,k'}u_\mathbf{k}v_\mathbf{k}^*u_{\mathbf{k}'}^*v_{\mathbf{k}'}$$

Since $|u_\mathbf{k}|^2+|v_\mathbf{k}|^2=1$

we can parameterize (real?):

$$u_k=\cos\theta_k\quad v_k=\sin\theta_k$$

$$
\begin{aligned}
E =& 2\sum_k \xi_k \sin^2\theta_k - \frac{g}{4V}\sum_{k,k'}\gamma_{k,k'}\sin 2\theta_k \sin 2\theta_{k'} \\
=& \sum_k \xi_k(1-\cos 2\theta_k) - \frac{g}{4V}\sum_{k,k'}\gamma_{k,k'}\sin 2\theta_k \sin 2\theta_{k'}
\end{aligned}
$$

suppose

$$
\gamma_{k,k'} = \gamma_k \times \gamma_{k'}
$$

variation (double summation)

$$
\frac{\partial}{\partial\theta_k}E = 2\xi_k \sin 2\theta_k - \frac{g}{V}\gamma_k \cos 2\theta_k \sum_{k'}\gamma_{k'}\sin 2\theta_{k'} = 0
$$

define

$$
\Delta_k = \frac{g}{2V}\gamma_k \sum_{k'}\gamma_{k'}\sin 2\theta_{k'}
$$

put it back

$$
\tan 2\theta_k = \frac{\Delta_k}{\xi_k}
$$

$$
\sin 2\theta_k = \frac{\Delta_k}{\sqrt{\Delta_k^2+\xi_k^2}}
$$

$$
\cos 2\theta_k = \frac{\xi_k}{\sqrt{\Delta_k^2+\xi_k^2}}
$$

put $\sin 2\theta_k$ back to the definition of gap, we can get the self-consistent gap function:

$$
\Delta_k = \frac{g}{2V}\sum_{k'}\gamma_k\gamma_{k'}\frac{\Delta_{k'}}{\sqrt{\Delta_{k'}^2+\xi_{k'}^2}}
$$

suppose $\Delta_k = \Delta\gamma_k$

$$
\begin{aligned}
\Delta\gamma_k =& \frac{g}{2V}\sum_{k'}\gamma_k\gamma_{k'}\frac{\Delta\gamma_k'}{\sqrt{\Delta_{k'}^2+\xi_{k'}^2}} \\
1 =& \frac{g}{V}\sum_k \frac{\gamma_k^2}{2\sqrt{\Delta_k^2+\xi_k^2}}
\end{aligned}
$$

$$
1 = \frac{g}{V}\sum_{\mathbf{k}}\frac{\gamma_k^2}{2\sqrt{\Delta_k^2+\xi_k^2}}\tanh\left(\frac{\sqrt{\Delta_k^2+\xi_k^2}}{2k_BT}\right)
$$

self consistent equation

appendix

$$
\begin{aligned}
&\langle \Psi_{\mathrm{G}} \mid c_{-k',\downarrow}c_{k',\uparrow} \mid \Psi_{\mathrm{G}}\rangle \\
=&\langle 0 \mid \left(u_{\mathbf{k}}^* + v_{\mathbf{k}}^*\hat{b}_{\mathbf{k}}\right)\left(u_{\mathbf{k}'}^* + v_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}'}\right) \mid \hat{b}_{k'} \mid \left(u_{\mathbf{k}'} + v_{\mathbf{k}'}\hat{b}_{\mathbf{k}'}^\dagger\right)\left(u_{\mathbf{k}} + v_{\mathbf{k}}\hat{b}_{\mathbf{k}}^\dagger\right) \mid 0\rangle \\
=&\langle 0 \mid (u_{\mathbf{k}}^*u_{\mathbf{k}'}^* + u_{\mathbf{k}}^*v_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}'} + v_{\mathbf{k}}^*u_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}} + v_{\mathbf{k}}^*v_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}}\hat{b}_{\mathbf{k}'}) \mid \hat{b}_{k'} \mid (u_{\mathbf{k}}u_{\mathbf{k}'} + u_{\mathbf{k}}v_{\mathbf{k}'}\hat{b}_{\mathbf{k}'}^\dagger + v_{\mathbf{k}}u_{\mathbf{k}'}\hat{b}_{\mathbf{k}}^\dagger + v_{\mathbf{k}}v_{\mathbf{k}'}\hat{b}_{\mathbf{k}'}^\dagger\hat{b}_{\mathbf{k}}^\dagger) \mid 0\rangle \\
=&\langle 0 \mid (u_{\mathbf{k}}^*u_{\mathbf{k}'}^* + u_{\mathbf{k}}^*v_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}'} + v_{\mathbf{k}}^*u_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}} + v_{\mathbf{k}}^*v_{\mathbf{k}'}^*\hat{b}_{\mathbf{k}}\hat{b}_{\mathbf{k}'}) \mid (u_{\mathbf{k}}v_{\mathbf{k}'} + v_{\mathbf{k}}v_{\mathbf{k}'}\hat{b}_{\mathbf{k}}^\dagger) \mid 0\rangle \\
=&u_{\mathbf{k}}^*u_{\mathbf{k}'}^*u_{\mathbf{k}}v_{\mathbf{k}'} + v_{\mathbf{k}}^*u_{\mathbf{k}'}^*v_{\mathbf{k}}v_{\mathbf{k}'} \\
=&u_{\mathbf{k}'}^*v_{\mathbf{k}'}(|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2) \\
=&u_{\mathbf{k}'}^*v_{\mathbf{k}'}
\end{aligned}
$$

double summation

$$
\begin{aligned}
&\frac{\partial}{\partial\theta_{k_0}}\left[\sum_{k,k'}\gamma_{k,k'}\sin 2\theta_k \sin 2\theta_{k'}\right]\\
=&\frac{\partial}{\partial\theta_{k_0}}\left[\gamma_{k_0,k_0}\sin^2 2\theta_{k_0}+\sum_{k\neq k_0}\gamma_{k_0,k}\sin 2\theta_{k_0}\sin 2\theta_k+\sum_{k\neq k_0}\gamma_{k,k_0}\sin 2\theta_k\sin 2\theta_{k_0}\right]\\
=&4\gamma_{k_0,k_0}\sin 2\theta_{k_0}\cos 2\theta_{k_0}+2\sum_{k\neq k_0}\gamma_{k_0,k}\cos 2\theta_{k_0}\sin 2\theta_k+2\sum_{k\neq k_0}\gamma_{k,k_0}\sin 2\theta_k\cos 2\theta_{k_0}\\
=&4\sum_k\gamma_{k,k_0}\sin 2\theta_k\cos 2\theta_{k_0}
\end{aligned}
$$